

Artificial Intelligence: Reshaping the Fields of Business and Law

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In the last few years, artificial intelligence, which has become an easily accessible tool used every day by everyone, is now one of “the most important technological advances in decades.” (Gates). AI has reshaped the foundations of both business and law. However, with such transformation comes a host of new opportunities and challenges. AI’s integration into these sectors offers significant potential for efficiency and productivity but also raises concerns about fairness, accountability, and legality. (West).

AI is revolutionizing how businesses operate and can “automate routine tasks, reduce operational costs, and increase productivity.” (Kutumbe). Machine learning algorithms are now able to detect consumer patterns, optimize supply chains, and even manage financial portfolios (Alicke). For example, Amazon uses AI for “forecasting, supply chain, and capacity planning, enabling efficient management of inventory for hundreds of millions of products.” (Amazon). Other companies like Netflix and Spotify use AI as well, and allow better recommendations for users. (Antonyuk). There are also many chatbots such as “Erica” used by Bank of America and financial services firms that are capable of handling thousands of customers at the same time, providing a 24/7 service, which reduces operational costs. (Bank of America).

Businesses are also starting to use AI for strategic insights. AI can analyze large data sets to help make decisions on marketing, product development, and risk management. (Bremen). In finance, some AIs use algorithms to build portfolios based on client risk profiles. (Ibid). However, reliance on AI raises questions about transparency as many systems operate as “black boxes,” meaning their decision-making processes are not clearly understood by humans. (Kosinski). For example, if an AI recommends a strategic move or denies a loan application, it can be difficult for stakeholders to understand these outcomes.

While some businesses benefit from the lower costs and better efficiency given by artificial intelligence, automation also threatens job displacement. A McKinsey report suggests

that “between 400 million and 800 million individuals could be displaced by automation and need to find new jobs by 2030.” (Manyika). Cashiers, truck drivers, journalists and factory workers are the roles with more risk of being completely replaced. (Liu). Additionally, biased AI systems can lead to discriminatory outcomes. In 2018, Amazon scrapped a secret AI hiring tool because it showed bias against female applicants. (Dastin). It was explained by the training received by the computer model which was observing patterns in resumes submitted to the company over a 10-year period, and most came from men. “Amazon's system taught itself that male candidates were preferable. It penalized resumes that included the word ‘women’s’” (Ibid).

Furthermore, AI has significant implications for cybersecurity. AI-driven cybersecurity tools can detect threats more quickly than traditional methods, and can help protect businesses from potential cyber-attacks, because it can “monitor, analyze, detect, and respond to cyber threats in real time, [...] and can also scan the entire network for weaknesses to prevent common kinds of cyber attacks.” (Sophos). Unfortunately, AI can also be used by people with malicious intent, and hackers can use it to develop sophisticated cyber attacks, making the AI-based security measures inefficient. Cyber attacks are not the only threat that we can have online. Scammers are now using AI to do a voice clone or a deep fake of someone, allowing them to create fake emergency situations on a voice call, or fake pictures that could damage someone’s reputation. (Orebaugh).

AI is similarly reshaping the legal practice by improving the speed and accuracy of research and documentation. Tools like ROSS Intelligence used natural language processing to respond to complex legal queries, but has been shut down since Thomson Reuters sued them in May 2020, “alleging the AI/legal research company unlawfully copied content from Thomson Reuters legal research platform Westlaw for the purpose of training its AI-based platform.” (bakerlaw). This automation by AIs allows lawyers to focus on higher-level strategies which can take a lot of time, rather than repetitive tasks, improving productivity and reducing costs for

clients. Some firms have even started using AI to predict case outcomes, allowing a better risk assessment and litigation planning. (Hallene).

AI also holds the potential to improve access to justice. Websites like DoNotPay or Cetient use artificial intelligence to help individuals to handle common legal tasks without needing to hire an attorney, making legal help more affordable and accessible. In the long term, the justice gap faced by low-income individuals can be narrowed if this technology is used appropriately. (Markovich). However, this raises ethical and regulatory concerns. If an AI system provides incorrect advice or information, who among the developer, the platform, or the user should be held liable. (Sahota). In addition, some Legal AI platforms often lack legal certification and “many jurisdictions have strict rules that restrict legal advice to licensed attorneys” (Smart advocate) and could constitute unauthorized practice of law. (Ibid).

Furthermore, AI “lacks human qualities such as empathy, contextual understanding, and moral reasoning” (Frankit) and these qualities are essential in legal decision-making. For example, in family law or criminal defense, some client’s situations require more than an algorithm from a computer. While AI can assist with legal tasks, “work product and conclusions reached by AI cannot replace human judgment and must be reviewed by lawyers for completeness and accuracy. (Duplantier).

One of AI’s primary legal challenges is accountability, particularly when AI can impact judicial decisions and transactions between different businesses. The case of State v. Loomis in 2016, which is one of the first major U.S. court cases to address AI, is a great example of this issue. The court used the COMPAS algorithm, which estimates the risk of recidivism of the offender. (Harvard Law). Loomis contested the ruling, “arguing that the court’s reliance on COMPAS violated his due process rights”. (Ibid). The court upheld the use of COMPAS but

emphasized it should not be the sole basis for decisions, a recognition of the legal and ethical concerns involved. (Ibid).

AI's dependency on large datasets also raises concerns about privacy. Clearview AI is a company that developed a powerful facial recognition app that allows users (primarily law enforcement) to upload a photo of a person and search a massive database of about 50 billion images scraped from the internet. (Clearview AI). The company faced multiple lawsuits for scraping these images from public websites without consent and “offering access to this database to private companies, wealthy individuals, and federal, state, and local law enforcement agencies.” (ACLU). The ACLU filed a lawsuit alleging Clearview violated the Illinois Biometric Information Privacy Act, which requires consent for the collection of biometric data. (ACLU).

AI's integration into intellectual property law raises new complex questions. AI systems such as Stable Diffusion, Midjourney, or DALL-E, can create original works, such as artworks, music, literature, and inventions, but they challenge the ideas of authorship and inventorship. In the case *Thaler v. Perlmutter* in 2022, the U.S. Copyright Office determined that works created by an AI do not qualify for copyright protection because “the Copyright Act requires all eligible works to be authored by a human being.” (Stanford Library). Additionally, inventions generated by AIs face similar hurdles. The U.S. Patent and Trademark Office have “ruled that an artificial intelligence cannot be listed as an inventor on a patent application.” (Rawls) emphasizing the requirement for human inventors. These legal questions have started debates about whether intellectual property laws should evolve to recognize AI as an author or maintain their human-centric approach. Current laws were not designed with AI in mind, and as technology advances, it is important to rethink “how to approach intellectual property rights. This could mean making new laws, changing current ones, or looking at old laws in different ways.” (Hess).

Moreover, addressing AI's environmental impact is increasingly necessary. The training of large AI models consumes vast amounts of energy, contributing significantly to carbon emissions. (Yao). For example, “researchers estimated that creating GPT-3, is the equivalent of 123 gasoline-powered passenger vehicles driven for one year.” (Saenko). Additionally, the growing demand for hardware and cooling systems for data centers, particularly in data-intensive industries, strains energy resources and its carbon footprint is expected to grow. However, AIs can have a positive impact on the environment when used wisely. They can enhance energy efficiency, reduce energy usage, assist in environmental monitoring, and support process and supply chain. (Yao). Also, researchers are exploring solutions such as the use of renewable energy in data centers, and the development of “green AI” which prioritize environmental sustainability during the development of artificial intelligences. (Bolón-Canedo). These efforts are really important not only to limit environmental damage, but also to ensure the viability of AI technologies in the world for the long-term.

In sum, artificial intelligence is transforming both business and law and offers new opportunities to increase efficiency and innovation, expand access to services, and reduce costs. AI is also capable of reshaping society by optimizing supply chains and improving cybersecurity to support legal research. However, these benefits come with concerns, ranging from job displacement and bias in both recruitment and legal contexts, to privacy risks and environmental issues. Cases like *State v. Loomis* and *Thaler v. Perlmutter* highlight very well how legal systems are starting to face the challenges given by AIs. While current laws struggle to keep pace with new technologies, the solution may not be to reject AI, but to regulate its use. Finally, AI is neither good nor bad, its impact will depend on how people choose to use it. If well integrated with ethical oversight, AI has the potential to serve as a powerful tool in both business and law.

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